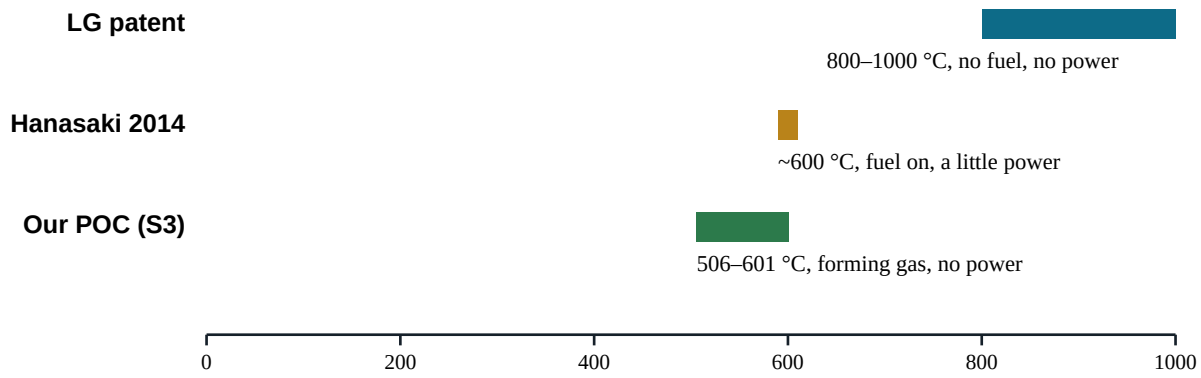
SOFC proof of concept · operating stages · 2026-10-29

Why stage names don't line up, and how to sort sensor readings anyway

The names are not the real problem. The meanings behind them are different. So we sort sensor readings by what the sensors actually show (six plain facts per minute). Then we translate those facts into each plant's stage names with a small lookup table.

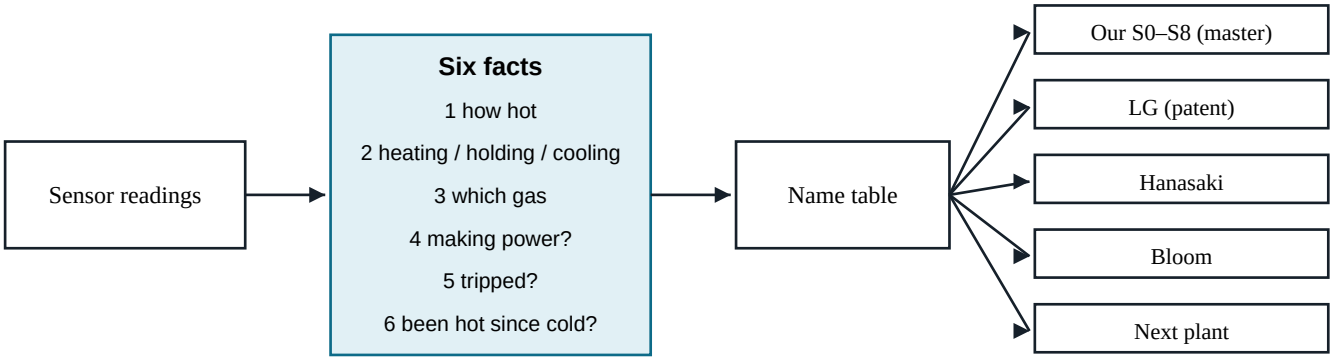
If we split readings by name, LG's "Hot standby" (800–1000 °C, no fuel) would be mixed with ours (about 500–600 °C) and with Hanasaki's (600 °C, still making a little power).

One word, three meanings



Stack temperature band each source calls "Hot standby". Ours is the middle 90 % of the 297 h filed as Hot standby (lowest 494 °C, highest 654 °C).

The fix: sort by facts, translate by table



The readings are sorted once. Each plant's names are a lookup table on top; a new plant adds rows, not new sorting.

What we tested on our 912 hours	Result
Fact combinations that occur	50
Minutes where the six facts alone give our own S0–S8 stage	90.6 %
Our hours matching an LG patent name exactly	47.9 %
Our hours LG has no name for (heating, cooling, trip, warm and off)	39.7 %

Where the LG names come from: LG patent US10797329, not the LG data-centre plant's own control screens. The plant's real list goes into the same table later.

If you remember one thing: Store the sensor readings once, tagged with what the plant was physically doing. Stage names from any plant are just a view on top.

Six sensor facts describe every minute

Each fact is something a sensor shows directly, so it means the same in any plant. Only the limits are set per plant.

Fact	Values	Read from	Our limits
1 How hot	Cold / Warm / Hot	Stack temperature	Cold ≤ 46 °C (room + 30 K); Hot ≥ 450 °C. Set per plant.
2 Trend	Rising / Held / Falling	Change over 20 min	Hot: ±5 °C/h. Below 450 °C: ±1 °C/h (natural cooling is slow).
3 Gas supply	Off / Air / Protective / Fuel	Gas and air flow controllers	Protective = forming gas or N2 purge; Fuel = natural gas.
4 Load	No load / Load / Rated	String current	Load ≥ 1 A; Rated ≥ 28.5 A.
5 Trip	Yes / No	Trip check	9 trips, 2.8 h.
6 Been hot since cold	Yes / No	Memory of facts 1–4	Yes once the stack held hot or made power since last cold.

Example: one minute, four names

Hot (511 °C), held (+0.4 °C/h), protective gas, 0 A, no trip, been hot → ours **S3 Hot standby**; LG **Hot standby mode**; Hanasaki **hot standby** (partial, theirs still makes power); Bloom **Steady state**.

How much of our own labelling the facts rebuild

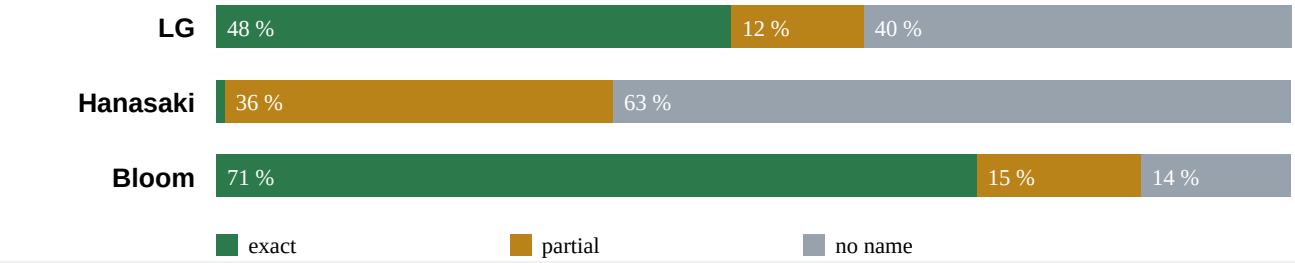
Same facts, different stage	Our stages (hours)	What tells them apart
Hot, rising, after being hot	S2 21.0 · S4 23.7	Whether the earlier hold was short or long
Hot, falling, after being hot	S7 46.6 · S4 16.5	What comes next: levels off, or goes to cold
Cold, steady, air on or off	S0 85.0 · S7 ~30	Our rule calls it cooling until the air stops
Warm, falling, plant off	S1 19.2 · S2 10.0 · S7 10.6	What came before

These four cases need the stage sequence, which our existing stage code already handles.

If you remember one thing: Temperature, trend, gas, current and trip, plus one memory flag, say what the plant was doing nine minutes in ten, whatever the plant calls it.

Our stages next to LG, Hanasaki and Bloom

We ran each outside name list over our 912 hours. **No outside list covers all our stages.**



Our stage (h)	LG patent	Hanasaki 2014	Bloom patent
S0 Cold, plant off (88.2)	Shut-down mode (87.2 h, exact) Cool-down transition (not named) (0.6 h, none)	Plant off (not named) (85.4 h, none) Heat-up (not named) (1.2 h, none) Cool-down (not named) (1.0 h, none)	Plant off (not named) (85.8 h, none) Start up operating mode (1.2 h, exact) Shut down operating mode (1.0 h, exact)
S1 Warm, plant off (21.0)	Cool-down transition (not named) (19.7 h, none) Heat-up transition (not named) (1.2 h, none)	Cool-down (not named) (19.2 h, none) Heat-up (not named) (1.2 h, none) Cold standby (but not a set hold) (0.5 h, partial)	Shut down operating mode (19.2 h, exact) Start up operating mode (1.2 h, exact) Plant off (not named) (0.5 h, none)
S2 Start-up (148.9)	Heat-up transition (not named) (90.0 h, none) Cool-down transition (not named) (26.2 h, none) Hot standby mode (set-point change) (22.9 h, partial)	Heat-up (not named) (117.4 h, none) Cool-down (not named) (18.6 h, none) Cold standby (but not a set hold) (10.6 h, partial)	Start up operating mode (96.4 h, exact) Steady state operating mode (22.9 h, partial) Shut down operating mode (16.7 h, exact)
S3 Hot standby (297.1)	Hot standby mode (292.9 h, exact) Hot standby mode (set-point change) (2.1 h, partial) Power-ready hot standby (2.0 h, exact)	Hot standby (but no power made) (293.0 h, partial) Heat-up (not named) (2.8 h, none) Cool-down (not named) (1.2 h, none)	Steady state operating mode (293.0 h, exact) Steady state operating mode (4.0 h, partial)
S4 Thermal adjustment (44.0)	Hot standby mode (set-point change) (40.5 h, partial) Hot standby mode (1.9 h, exact) Power-ready hot standby (1.4 h, exact)	Heat-up (not named) (25.2 h, none) Cool-down (not named) (16.7 h, none) Hot standby (but no power made) (2.0 h, partial)	Steady state operating mode (41.9 h, partial) Steady state operating mode (2.0 h, exact)
S5 Part load (4.4)	Normal operating mode (4.4 h, exact)	Operation (4.4 h, exact)	Steady state operating mode (4.4 h, exact)

Our stage (h)	LG patent	Hanasaki 2014	Bloom patent
S7 Shut-down (307.1)	Cool-down transition (not named) (220.4 h, none) Hot standby mode (set-point change) (46.6 h, partial) Shut-down mode (37.9 h, exact)	Cool-down (not named) (253.0 h, none) Plant off (not named) (29.8 h, none) Cold standby (but not a set hold) (21.3 h, partial)	Shut down operating mode (206.4 h, exact) Steady state operating mode (46.7 h, partial) Plant off (not named) (39.0 h, none)
S8 Trip (2.8)	Trip (not named in LG) (2.8 h, none)	Shutdown (emergency) (2.8 h, exact)	Shut down operating mode (2.8 h, partial)

Top three names per stage, hours, and match grade (exact / partial / none).

What stands out

• **LG fits our still, hot stages almost perfectly**: S0 → Shut-down mode 99 %, S3 → Hot standby mode 99 %, S5 → Normal operating 100 %. LG's Power-ready hot standby = our Pre-load (3.4 h).
• **LG has no name for heating up or cooling down**: most of S2 and S7, about 40 % of hours, where sensors move most.
• **No trip stage in LG**; only Hanasaki's emergency shutdown matches our S8.
• **Hanasaki's hot standby still makes power**, so it is only a partial match.
• **Bloom's three modes are too coarse**: S3, S4 and S5 all become steady state.

If you remember one thing: Keep our S0–S8 as the master list. Outside lists become translations on top, and each shows clearly where it has no name.

What to do next

The method is built and tested on our data. To bring in the LG data-centre plant, **we need that plant's own stage list**.

How to separate sensor readings by stage from now on

1. Tag every minute of every plant with the six facts (our 912 h fall into 50 combinations).
2. Compute sensor statistics per fact combination; these compare across plants.
3. Report in a plant's own words by joining its name table, or in ours with S0–S8.

Decisions for you

	Question	Suggested answer
A	Which list is the master?	Our S0–S8, the only list with a name for every hour.
B	How is "hot" set for a hotter plant (LG 800–1000 °C)?	Per plant, from its own standby temperature, not our 450 °C.
C	Stages a plant has no name for (LG: heat-up, cool-down, trip)?	Keep our name, mark "not named in LG"; never force the nearest LG name.
D	What to ask LG and other plants for	Control-screen stage names, entry and exit conditions, and whether a stage tag is logged.

Where the work is: data_analysis/stage_crosswalk.py; outputs in data_analysis/stage_crosswalk/. Adding a plant = adding rows to its table. No existing stage rule changed.

If you remember one thing: The missing piece is the LG plant's own stage list with its conditions. Once we have it, it slots into the table and its readings sort the same way as ours.